Vinayak

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Education

University of Massachusetts, Amherst, USA — GPA: 3.94/4.0
M.S./Ph.D. in Computer Science — 2023 - Present
Advisor: Andrew McGregor

Indraprastha Institute of Information Technology, Delhi, India — GPA: 8.99/10.0
B.Tech. (with Honors) in Computer Science & Engineering — 2017 - 2021

Research Interests

Structural Graph Theory, Metric Embeddings

Publications

[1] An La, Hung Le, Shay Solomon, Cuong Than, **Vinayak**, Shuang Yang, Tianyi Zhang. *Optimal Bounds for Spanners and Tree Covers in Doubling Metrics.* In Proceedings of the 42nd International Symposium on Computational Geometry (SOCG) 2026. https://doi.org/10.48550/arXiv.2508.11555.

[2] Kacper Kluk, Hung Le, Wojciech Nadara, Marcin Pilipczuk, Hector Tierno, **Vinayak**. *A Polynomial Coreset for Furthest Neighbor in Planar Metrics.* To appear in the Proceedings of the 67th IEEE Symposium On Foundations Of Computer Science (FOCS) 2026. https://doi.org/10.48550/arXiv.2603.28100.

[3] Hsien-Chih Chang, Jonathan Conroy, William Eliot, Hung Le, **Vinayak**. *Peeling Half the Onion: Embedding k-Outerplanar Graphs into ℓ_1 and Trees with a Polynomial Distortion (in k).* To appear in the Proceedings of the 2027 Annual ACM-SIAM Symposium on Discrete Algorithms (SODA).

Experience

Oracle — Bangalore, India
Member Technical Staff — July - Nov 2021
Worked as a software developer in the Cloud Infrastructure Networks team.

Selected Coursework

Undergraduate: Approximation Algorithms, Randomized Algorithms, Complexity Theory, Computability Theory, Quantum Computing, Linear Optimization, Combinatorics, Advanced Linear Algebra, Number Theory.
Graduate: Randomized Algorithms, Algorithmic Game Theory, Artificial Intelligence, Quantum Information Systems.

Academic Service

- Shadow Reviewer, SODA 2026.
- Subreviewer, SoCG 2026.
- Subreviewer, COCOON 2026.

Workshops & Conferences

- 42nd Symposium on Computational Geometry — Rutgers University, USA, 2026
- DIMACS Workshop on Hardness of Approximation in P. — Rutgers University, USA, 2025
- DIMACS Tutorial on Fine-grained Complexity. — Rutgers University, USA, 2024
- ACM Winter School on Algorithms for Big Data. — IMSc Chennai, India, 2020
- ACM Summer School on Geometric Algorithms. — NISER Bhubaneswar, India, 2019

Teaching Experience

- Teaching Assistant, COMPSCI 611: Advanced Algorithms (Spring 2024, Spring 2025, Fall 2026)
- Teaching Assistant, COMPSCI 601: Computation Theory (Fall 2025)
- Mentor, UMass Early Research Scholars Program (Spring 2026)

Skills

- **Languages:** C, C++, Python, Java, MATLAB.
- **Tools/Frameworks:** LaTeX, Qiskit, PyTorch, Git.

Honors and Awards

- Top Quartile, Simon Marais Mathematics Competition (2020).
- Qualified for ACM-ICPC Amritapuri Regionals 2019 (top 100).